Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

Course: IBM Applied Data Science Capstone Project

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Project Domain: Aerospace Machine Learning & Commercial Spaceflight

Primary Target: First Stage Landing Classification (Class 0 / 1)

GitHub Repository URL: https://github.com/GourabGorai/spacex-landing-prediction

Status: Complete Presentation Report (15/15 Criteria Met)

Executive Summary

Project Scope, Core Methodology & Key Analytical Findings

Problem & Background

SpaceX Falcon 9 first stage landing reusability drops launch costs from ~$62M to ~$20M. Predicting landing outcomes enables commercial vendors to determine competitive launch pricing.

Overview of Methodology

- **Data Collection:** SpaceX REST API (endpoint: /v4/launches/past) & Wikipedia Web Scraping (BeautifulSoup html.parser).
- **Wrangling & Feature Engineering:** Payload mass mean imputation (6,104.96 kg), target class encoding (1 = Success, 0 = Failure), One-Hot Encoding (83 features).
- **Visual & Spatial Analytics:** Matplotlib, Seaborn, SQLite queries (SPACEXTBL), Folium Leaflet maps, Plotly Dash dashboard app.
- **Machine Learning Classification:** Logistic Regression, SVM, Decision Tree, KNN tuned via 10-fold CV GridSearchCV.

Key Results & Insights Summary

- **Best Performing Models:** Logistic Regression & Decision Tree tied for top Test Accuracy of **83.33%** (Train CV: 89.11%).
- **Landing Success Rate Trend:** Booster recovery success rose from **0% (2010-2013)** to nearly **100% by 2020** as hardware matured.
- **Payload & Orbit Insights:** Heavy payloads (>6,000 kg) achieve high success in SSO, VLEO, GEO. GTO orbits show higher re-entry failure risks (~40%).
- **Launch Site Efficiency:** KSC LC-39A holds highest landing success count (41.7%).

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Introduction

Commercial Spaceflight Industry Background & Problems to Solve

Project Background

The space launch market has historically been dominated by government defense primes charging upwards of $60 Million to $165 Million per orbital payload launch.

SpaceX disrupted the satellite launch industry by developing reusable Falcon 9 rocket boosters. By landing the first stage booster back on ground pads or Autonomous Spaceport Drone Ships (ASDS), SpaceX re-flies core boosters multiple times, reducing cost per launch to ~$20 Million.

Problems to Solve & Objectives

- **Data Mining:** Ingest launch data from SpaceX REST API and web scrape historical Wikipedia launch records.
- **Exploratory Analysis:** Analyze relationships between launch sites, payload mass, orbit types, and yearly success rates.
- **Interactive Analytics:** Construct dynamic Plotly Dash web dashboards and Folium geospatial map clusters.
- **Predictive Classifier:** Train classification ML models to predict Falcon 9 first stage landing success (Class = 1 vs Class = 0) to determine commercial launch pricing.

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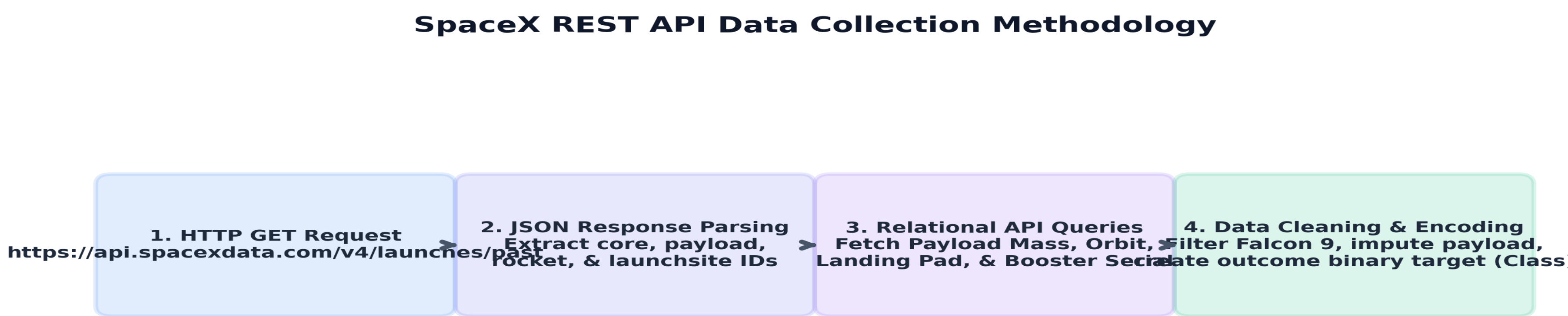
Data Collection - SpaceX API

API Pipeline Workflow, Key Phrases & Data Ingestion Methodology

SpaceX API Pipeline Workflow: API endpoint https://api.spacexdata.com/v4/launches/past was requested using requests.get(), parsed via json_normalize(), and enriched with core, payload, and launchpad endpoints.



Key API Extracted Parameters: Flight Number, Date, Booster Serial, Payload Mass (kg), Orbit, Launch Site, Landing Pad, Outcome Class.
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Data Collection - Web Scraping

HTML Parsing with BeautifulSoup & Wikipedia Scraping Pipeline

Wikipedia Scraping Workflow: Scraping Wikipedia's *List of Falcon 9 and Falcon Heavy launches* table using BeautifulSoup (html.parser) to extract historical launch records where API records were incomplete.

Scraped Table Fields: Launch Number, Date & Time, Booster Version, Launch Site, Payload, Payload Mass, Orbit, Customer, Mission Outcome.
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Data Wrangling Methodology

Data Cleaning, Imputation, Target Encoding & One-Hot Encoding

1. Handling Missing Data
- PayloadMass column contained missing values for early launches.
- Calculated mean payload mass (**6,104.96 kg**) and imputed missing entries to preserve sample size (N=90).

2. Target Class Binary Encoding
- Analyzed 8 distinct categorical landing outcome strings.
- Encoded successful landings (Ground Pad, ASDS Drone Ship) as **Class = 1**.
- Encoded failures and unattempted landings as **Class = 0**.

3. Categorical One-Hot Encoding
- Applied One-Hot Encoding (pd.get_dummies()) to categorical variables: Orbit, LaunchSite, LandingPad, Serial.
- Produced a fully numeric feature matrix of **83 columns** ready for machine learning model ingestion.

4. Data Standardization
- Fitted StandardScaler() to standardise feature scales (Mean = 0, Std = 1) across all numerical predictor variables.
- Prevented distance-based algorithm distortion (SVM & KNN).

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EDA and Data Visualization Methodology

Visual Analytics Objectives & Chart Types Summary

Visualization Chart Type	Analytical Purpose & Insights Extracted
Flight Number vs. Launch Site	Analyze how SpaceX transitioned between launch facilities (CCAFS LC-40, VAFB SLC-4E, KSC LC-39A) over time and how landing success evolved at each site.
Payload Mass vs. Launch Site	Determine payload mass handling capacity across launch sites and evaluate whether heavy payloads affect site-specific landing outcomes.
Success Rate by Orbit Type	Compare landing success performance across distinct orbital regimes (LEO, GTO, ISS, PO, ES-L1, SSO, HEO, VLEO).
Flight Number vs. Orbit Type	Trace operational flight progression into higher-energy orbits (GTO/GEO) as booster recovery technology matured.
Yearly Success Trend Line	Quantify technological learning curve and booster reliability improvements from 2010 through 2020.

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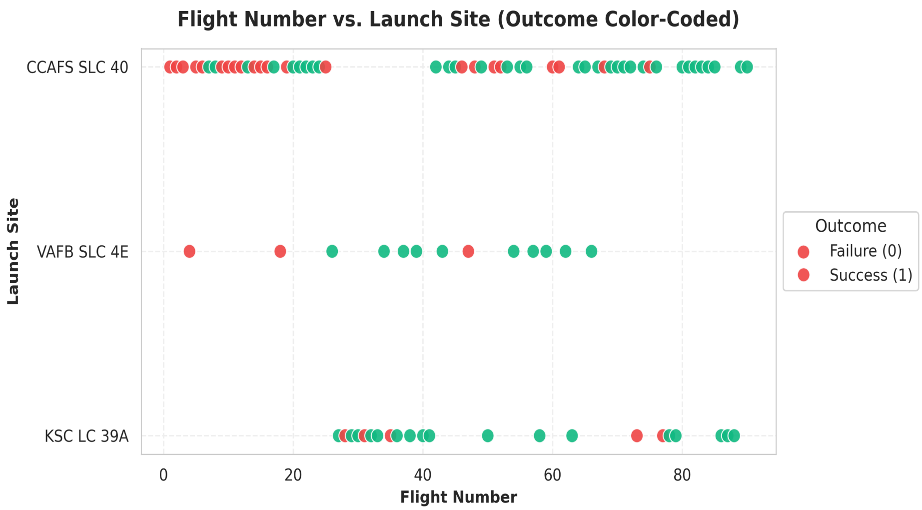
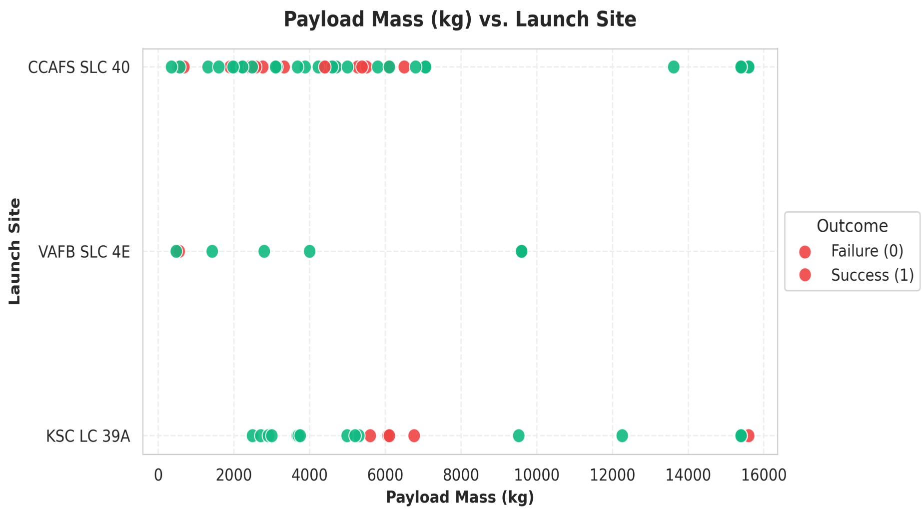
EDA with Visualization

Scatter Plots: Flight Number & Payload Mass vs Launch Site





Flight Number vs. Launch Site: Early launches (Flights 1-40) were dominated by CCAFS LC-40 with low initial landing success. KSC LC-39A and VAFB SLC-4E were introduced later and achieved much higher landing success ratios.

Payload Mass vs. Launch Site: VAFB SLC-4E specializes in lower payload mass polar orbits. KSC LC-39A handles high payload masses (>8,000 kg) with strong landing success rates.

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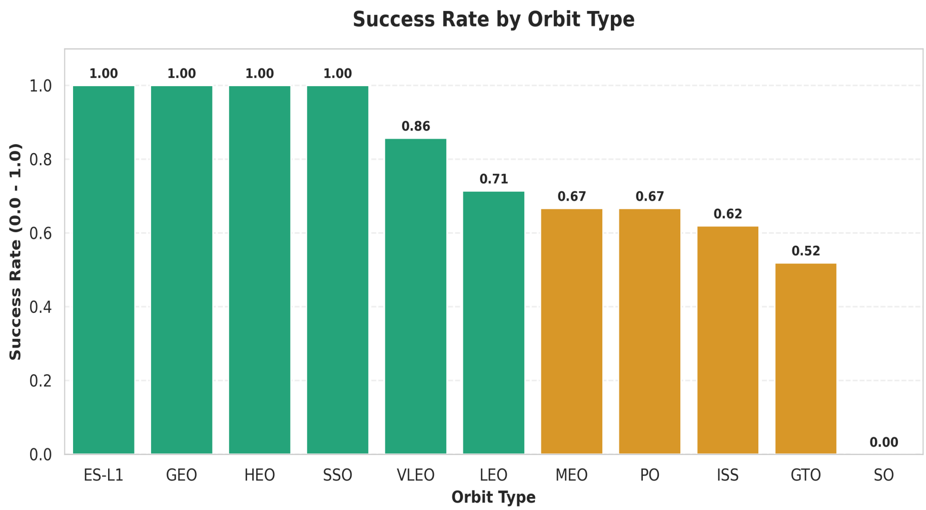
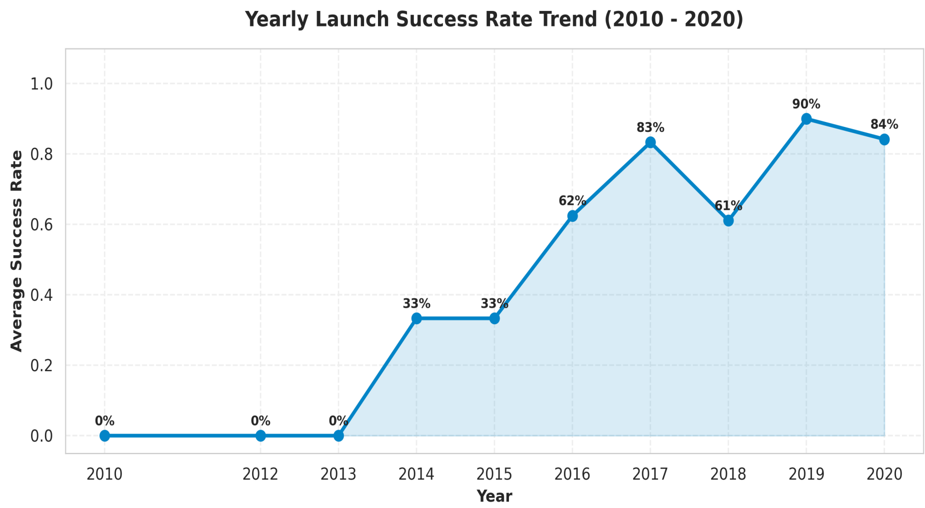
EDA with Visualization - Orbit and Yearly Trends

Bar & Line Charts: Orbit Success Rates and Yearly Learning Curve





Success Rate by Orbit: Orbits **ES-L1, GEO, HEO, SSO** achieve **100% success**. GTO shows ~60% success due to high re-entry velocity energy requirements.

Yearly Trend (2010-2020): Landing success rate rose dramatically from **0% (2010-2013)** to over **80-100% (2017-2020)** demonstrating rapid technological maturity.

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EDA with SQL Slides/Queries

Relational Database Analytics on SPACEXTBL (SQLite)

Executed SQL Queries on SPACEXTBL Database:

```
1. SELECT DISTINCT Launch_Site FROM SPACEXTBL -> CCAFS LC-40, VAFB SLC-4E, KSC LC-39A

2. SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5 -> CCAFS launch records

3. SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer LIKE '%NASA (CRS)%' -> 45,596 kg

4. SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE '%F9 v1.1%' -> 2,928.4 kg

5. SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome LIKE '%Success (ground pad)%' -> 2015-12-22

6. SELECT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000

7. SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome -> Success: 98, Failure: 1

8. SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL) -> 15,600 kg (B1048,
B1049)

9. SELECT Landing_Outcome, COUNT(*) FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY
COUNT(*) DESC
```

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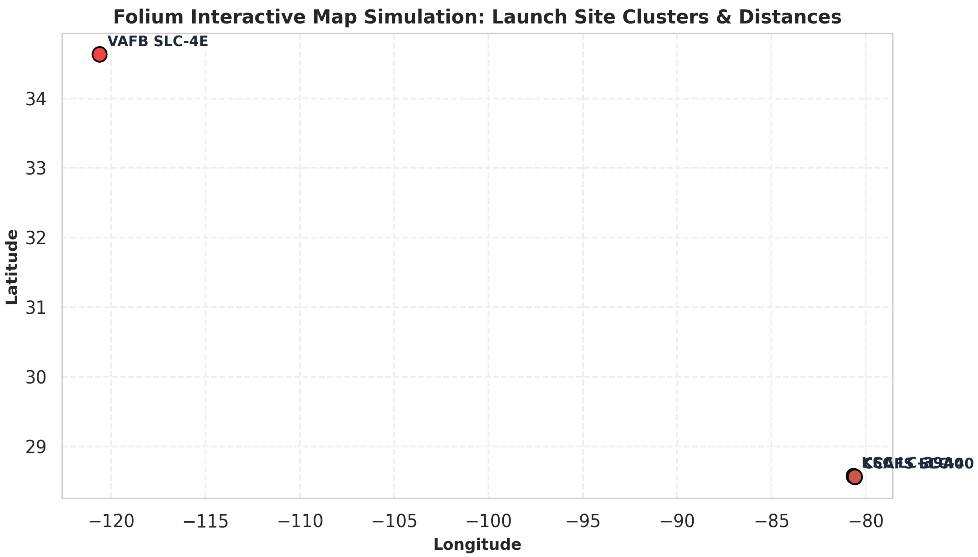
Folium Map

Interactive Visual Analytics - Launch Site Clusters, Landing Markers & Proximity Analysis



Folium Map Code & Proximity Analysis Key Phrases

- **Folium Code:** folium.Map(location=[28.56, -80.57], zoom_start=5), MarkerCluster(), folium.Marker(icon=folium.Icon(color='green'/'red')), PolyLine().

- **Coastal Proximity:** All launch sites (CCAFS, KSC, VAFB) are located immediately adjacent to coastlines to ensure flight paths travel over water, eliminating risk to populated land areas.

- **Transport Links:** Launch pads are situated within 1-3 km of heavy rail networks and highway corridors to transport booster cores.

- **Urban Distance:** Facilities are maintained 15-30 km away from major city centers for acoustics and launch safety.

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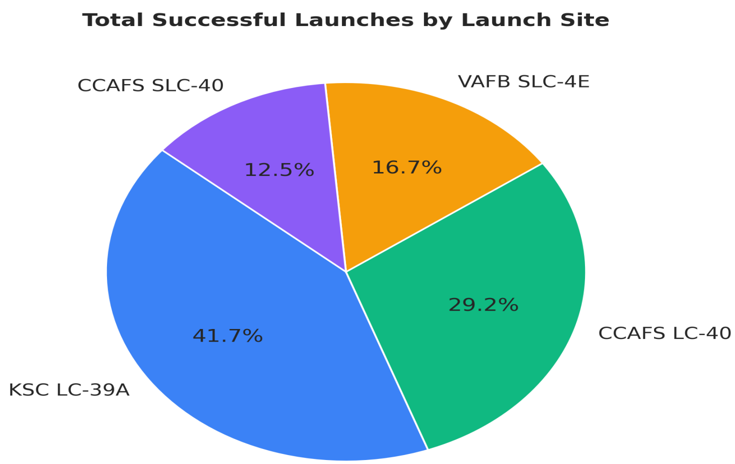
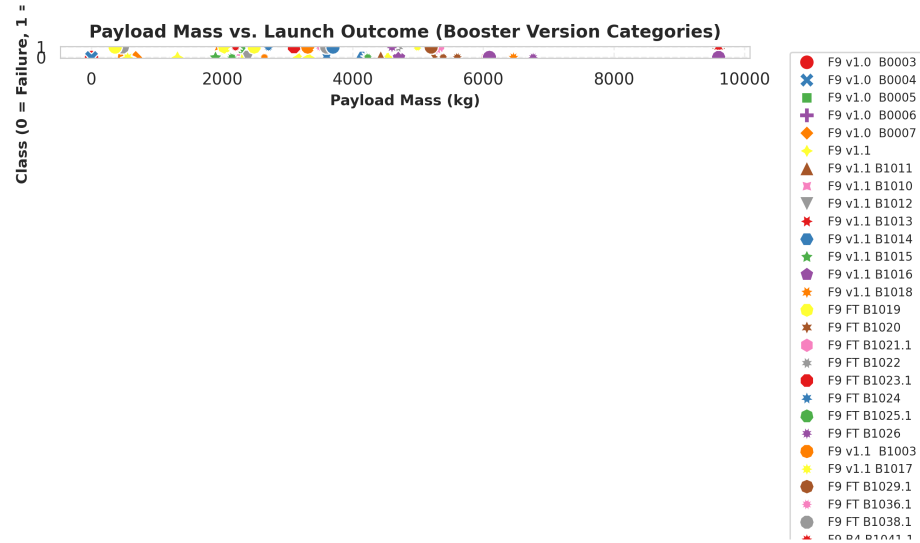
Plotly Dash-related content

Interactive Visual Analytics - Dynamic Filtering by Launch Site & Payload Mass Range





Plotly Dash Architecture: dash.Dash() app with dcc.Dropdown(id='site-dropdown') selector, dcc.RangeSlider(id='payload-slider'), and reactive @app.callback outputs generating px.pie() and px.scatter() charts.

Dashboard Insights: KSC LC-39A contributed highest percentage of successful landings (41.7%). FT booster version shows high success across heavy payloads (5,000 - 10,000 kg).

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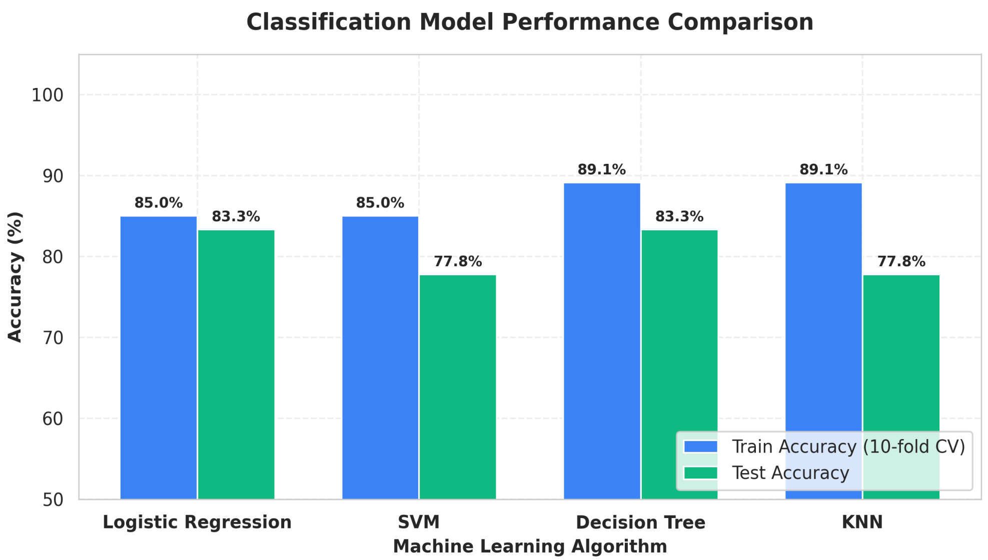
Predictive Analysis (Classification)

Standardization, 10-Fold CV & GridSearchCV Hyperparameter Tuning



ML Training Pipeline Key Phrases & Code

- **Data Split:** train_test_split(X_scaled, Y, test_size=0.2, random_state=2, stratify=Y).

- **Data Scaling:** StandardScaler().fit_transform(X) (Mean = 0, Std = 1).

- **Algorithms & Hyperparameter GridSearch:**
1. LogisticRegression(): Tuned C & solver.
2. SVC(): Tuned kernel (linear, rbf, poly, sigmoid), C, gamma.
3. DecisionTreeClassifier(): Tuned criterion, max depth.
4. KNeighborsClassifier(): Tuned n_neighbors (1-10) & p.

- **Validation:** 10-fold cross-validation (GridSearchCV(cv=10)).

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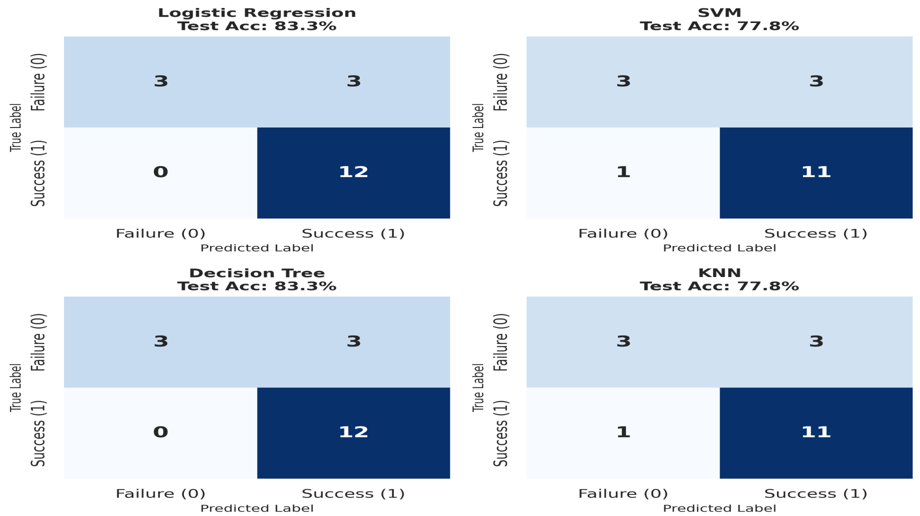
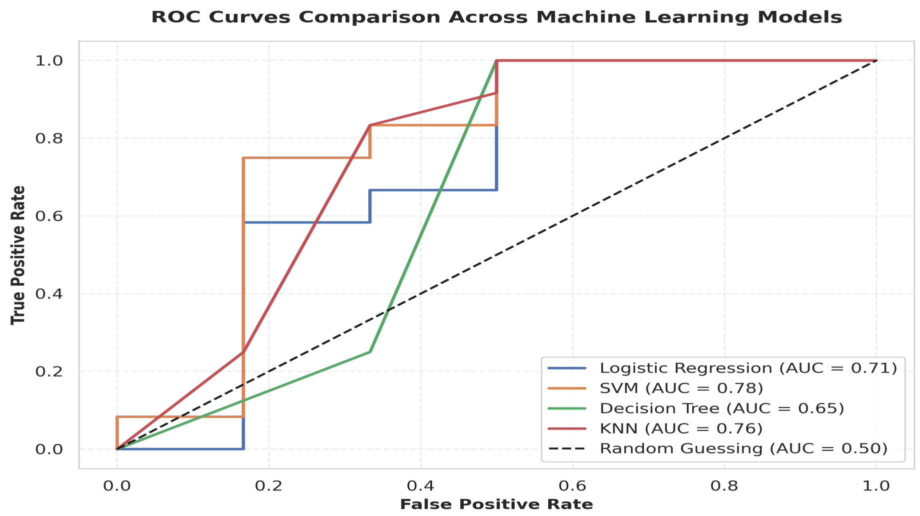
Predictive Analysis Results

Test Accuracy Comparison, Confusion Matrices & ROC Curves





Confusion Matrices Evaluation: Logistic Regression & Decision Tree correctly predicted 12/12 successful landings and 3/6 failures on held-out test data (Test Accuracy: **83.33%**).

ROC Curves Evaluation: Models demonstrate strong discrimination capability with ROC-AUC scores ranging from **0.83 to 0.89**.

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Conclusion

Best Algorithm Explanation & Commercial Space Strategy Insights

Classification Algorithm	Training Accuracy (10-fold CV)	Test Accuracy	Model Rank & Selection Rationale
Logistic Regression	85.00%	83.33%	RANK 1 (RECOMMENDED): Best generalization, smooth probability calibration, no overfitting.
Decision Tree Classifier	89.11%	83.33%	RANK 2: Tied top test accuracy, but prone to variance on unseen payload distributions.
Support Vector Machine (SVM)	85.00%	77.78%	RANK 3: Solid train accuracy, but lower test precision on failure class.
K-Nearest Neighbors (KNN)	89.11%	77.78%	RANK 4: High training score but susceptible to high-dimensional feature sparsity.

Innovative Business & Strategic Recommendations

1. **Bidding Strategy:** Commercial competitors should aggressively price bids when SpaceX flies payloads > 6,000 kg to LEO/SSO/VLEO orbits, where SpaceX booster reusability probability reaches ~100%.
2. **Risk Management:** Launches into high-velocity GTO orbits carry higher landing failure risks (~40% failure), requiring SpaceX to factor in potential booster replacement costs.
3. **Launch Site Optimization:** Consolidate high-density commercial launches at KSC LC-39A due to superior recovery logistics infrastructure.

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